



UNIT I

Errors Analysis

Syllabus :

Significant figures, round-off, precision and accuracy, approximate and true error, truncation error and Taylor series, machine epsilon, data uncertainties, error propagation, importance of errors in computer programming

Syllabus Topic : Significant Figures

1.1 Significant Figures

Review Question

Q. What is significant Figures ? Explain with suitable example.

- If a number is written so that there is no figures to the left of the decimal, then the number is said to be in its normalised floating point form.
- If a number is expressed in its normalized floating point form, then the numbers to the right of the decimal are referred to as the number of significant figures.
- For example :
 - (i) The number $\frac{1}{2}$. It can be written as 0.5×10^0 and has one significant figure.
 - (ii) The number $\frac{1}{3}$. It can be written as 0.3333×10^0 . Then this is a fraction upto four significant figures.
 - (iii) The number 126.4×10^2 . We first normalise it. i.e. 0.1264×10^5 . It has 4 significant figures.
 - (iv) The number 0.000372×10^1 .
In normalised form it is $0.372 \times 10^1 \times 10^{-3} = 0.372 \times 10^{-2}$. It has three significant figures.
- The quantities such as π , e or $\sqrt{7}$ represent specific quantities, they can not be expressed exactly by a limited number of figures. For example the quantity.

$$\pi = 3.145926535\ 89793238462643\ldots$$

Definition of Round off error

As computers can only retain a finite number of significant figure, such numbers can never be represented exactly. The omission of the remaining significant figure is called round off error.

Algebra of Normalised Floating Point Numbers

1. Addition of Two Normalised Floating Point Numbers

- Convert the numbers into normalised form.
- Make the exponents equal and add.

Example 1.1.1

Add the following two floating point numbers :

(a) 0.471923×10^7 , (b) 0.783329×10^7 .

Solution :

Here the exponents are equal and the numbers are in normalised form. We add directly.

$$\begin{array}{r} 0.471923 \times 10^7 \\ + 0.783329 \times 10^7 \\ \hline = 1.255252 \times 10^7 \end{array}$$

Hence, normalising we get,

0.1255252×10^8 is the solution

...Ans.

Example 1.1.2

Add the following two floating point numbers :

243.31947×10^{17} and 32.1698×10^{14} .

Solution : First we normalise the numbers.

$$0.24331947 \times 10^{20} \text{ and } 0.321698 \times 10^{16}$$

To add, we make exponents equal; i.e.

$$0.24331947 \times 10^{20} \text{ and } 0.0000321698 \times 10^{20}$$

Now we add them, i.e.

$$\begin{array}{r} 0.24331947 \times 10^{20} \\ + 0.0000321698 \times 10^{20} \\ \hline 0.2433516398 \times 10^{20} \end{array}$$

$0.2433516398 \times 10^{20}$ is the solution.

...Ans.

Example 1.1.3

Add the following two floating point numbers :

274.39242×10^{30} and 33.1122×10^{28} .

Solution : First convert them to normalised form :

$$\text{i.e. } 0.27439242 \times 10^{33} \text{ and } 0.331122 \times 10^{28}$$

To add, we make the exponents equal;



$$\begin{array}{r} \text{i.e. } 0.27439242 \times 10^{33} \text{ and} \\ + 0.00000331122 \times 10^{33} \\ \hline 0.27439573122 \times 10^{33} \end{array}$$

$0.27439573122 \times 10^{33}$ is the solution. ...Ans.

2. Subtraction of Two Normalised Floating Point Numbers

- Convert the numbers into normalised form.
- Make the exponents equal and subtract.

Example 1.1.4

Subtract the following floating point numbers : 0.3942163×10^7 , 0.24132×10^7

Solution : Since the numbers are in normalised form and exponents are equal;

$$\begin{array}{r} \text{hence } 0.3942163 \times 10^7 \text{ and} \\ - 0.24132 \times 10^7 \\ \hline 0.1528963 \times 10^7 \end{array}$$

0.1528963×10^7 is the solution ...Ans.

Example 1.1.5

Subtract the following floating point numbers : 416.21943×10^{18} , 37.9341×10^{19}

Solution : We express them in the normalised form : $0.41621943 \times 10^{21}$ and 0.379341×10^{21}

Since the exponents are same; we subtract them.

$$\begin{array}{r} 0.41621943 \times 10^{21} \\ - 0.379341 \times 10^{21} \\ \hline 0.03687843 \times 10^{21} \end{array}$$

0.3687843×10^{20} is the solution ...Ans.

Example 1.1.6

Subtract the following floating point numbers : 17.314298×10^{20} , 4.193247×10^{16} .

Solution : In normalised form, the numbers are

$$0.17314298 \times 10^{22} \text{ and } 0.4193247 \times 10^{17}$$

We make exponents equal and subtract them

$$\begin{array}{r} 0.173142980000 \times 10^{22} \\ - 0.000004193247 \times 10^{22} \\ \hline 0.173138786753 \times 10^{22} \end{array}$$

$0.173138786753 \times 10^{22}$ is the solution ...Ans.

3. Multiplication of Two Normalised Floating Point Numbers

- Convert the numbers into normalised form.
- Make the exponents equal and subtract.

Example 1.1.7

Multiply the floating numbers 47.31834×10^{15} and 3.1942×10^{12}

Solution : First we normalise the numbers :

$$0.4731834 \times 10^{17} \text{ and } 0.31942 \times 10^{13}$$

and take the product of numbers and add the indices.

$$\begin{array}{r} \text{i.e. } 0.4731834 \times 10^{17} \\ \times 0.31942 \times 10^{13} \\ \hline 0.151144241 \times 10^{30} \end{array}$$

$0.151144241 \times 10^{30}$ is the solution. ...Ans.

4. Division of Two Normalised Floating Point Numbers

- Convert the numbers into normalised form.
- Make the exponents equal and subtract.

Example 1.1.8

$$\text{Evaluate } \frac{317.23 \times 10^{33}}{32.17 \times 10^{18}}$$

Solution : Normalising the numbers, we get

$$\begin{aligned} &= \frac{0.31723 \times 10^{36}}{0.3217 \times 10^{20}} \\ &= 0.986105 \times 10^{16} \end{aligned}$$

0.986105×10^{16} is the solution ...Ans.

The above discussion, has one significant aspect in computations and that is "Significant Figures." The significant figures of a number are those, which can be used with more confidence (i.e. no ambiguity or with definiteness). The significant figures are important for us, because of the following reasons :

- Numerical methods, result in approximate values. So we must have a criteria to specify how confident are we in the approximated values.
- Some quantities, as mentioned earlier like π , represent some value, but infinite. Thus we need to choose only the significant figures from these (as computer can't manipulate finite numbers).

While doing so, one may leave out important figures also. This introduces the errors in the result. Our next section talks about the error.

1.2 Errors

Computational techniques lead to approximate solutions. Approximate solutions involve error. The main aim of computational technique is to reduce the error to minimum and to obtain better approximation. Hence we have to define various types of errors. We have to discuss the propagation of errors in the computational techniques.

Errors can be introduced in three ways

- Mistakes made by the person, carrying out the calculations.
 - The use of the inaccurate formula.
 - The use of inaccurate data, including the effects of round-off.
- (i) Actually all errors made under the heading shouldn't be there. But if an operator is using a machine that has developed a fault which he cannot detect. Mistakes commonly made by a human operator occur when copying and when doing mental arithmetic. Two common copying errors are :
- The reversal of two digits e.g. writing down 236721 instead of 263721 and
 - The repetition of the wrong digit, e.g. 233721 instead of 223721.

- (ii) An inaccurate formula may arise due to chopping off an infinite series after a finite number of terms. For example, when we obtain Simpson's rule by using Taylor's series, we expand $f(x_0 - h)$ and $f(x_0 + h)$ in powers of h but all terms involving powers greater than the second are dropped. We can now formally define the "error" as the quantity which is equal to the difference between the true value and approximated value, i.e.

$$e_t = \text{True value} - \text{Approximated value}$$

This e_t refers to the "true error." However note that here we have not talked about order of magnitude of the value being measured. For example, an error in gm (grams) is much more significant if we are measuring a gold, than in vegetable. To account the magnitude, we can normalise the error to the true value as,

$$e_{nt} = \frac{e_t}{\text{true value}}$$

where, $e_{nt} \Rightarrow$ normalised (true fractional relative error)

$e_t \Rightarrow$ true error

Expressing this quantity in %, we get true percentage relative error,

$$E = e_{nt} \times 100$$

Now let us consider absolute error in algebraic operations such as addition, subtraction, multiplication and division.

1.2.1 Types of Errors

Review Question

Q. Define and explain types of errors.

Basically there are five types, which are encountered in performing numerical computations.

These are as follows :

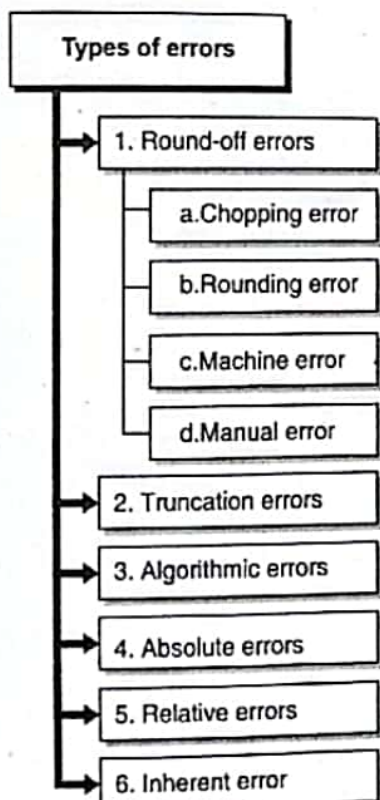


Fig. C1.1 : Types of errors

Syllabus Topic : Round-off Error

1.2.1.1 Round off Error

Review Question

Q. Explain round off error.

- As discussed earlier, there will be errors while performing arithmetic operations using normalised, floating point numbers are called round off errors.
- Some rational numbers require an infinite number of digits after the decimal point. Examples are $\frac{5}{3}$ and $\frac{22}{7}$ where we have the fixed point representations 1.666 and 3.1428571428571 In the second case we observe that the sequence of digits 142857 repeats indefinitely. In practice, we can work only with numbers containing a finite number of digits and we can approximate $\frac{22}{7}$ by say, the number 3.142857 with error 0.00000014 and we say that 3.142857 is $\frac{22}{7}$ correct (rounded) to six decimal place accuracy and 0.00000014..... is known as rounded error or round off error.
- In contrast, 1.66667 is $\frac{5}{3}$ rounded to six decimal places; here we have rounded up rather than rounded down because 1.6667 is closer to $\frac{5}{3}$ than 1.666666. The process of rounding should be compared with that of chopping. In the latter we simply remove all digits after a specified number of them.
- Note that rounding errors may be incurred even when it is possible to express a decimal number using a finite number of digits. Thus, an exact number in which we are interested might be 187.452963 but, because of limitations imposed by computing machinery, suppose we can work only with six digits available for the mantissa. The approximation to this number is 0.187453×10^3 and a rounding error of 0.37×10^{-4} has been incurred.
- The computers or calculators can store only a limited digits of a number. This neglects the further or higher digits as we have seen with the value of π .

There are four types of round-off errors. They are as follows :

→ a. Chopping error

In this a number is chopped after required number of digits without considering values of deleted digits.

e.g. 2.34567 after chopping last two digits the number becomes 2.345.

→ b. Rounding error

In this a number is rounded i.e. the last retained digit is changed based on value of deleted digits.

e.g. 2.34567 after removing last two digits, the number becomes 2.346.

→ c. Machine error

Any microprocessor, and memory in a computer system have limited number of storage space for each number e.g. 8 bit, 16 bit, 32 bit, 64 bit etc. out of which some bits are used to store exponent and rest to store mantissa.

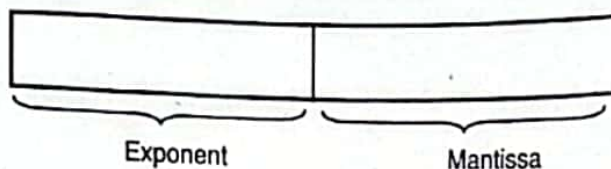


Fig. 1.2.1

If the result is having more number of digits in mantissa or (in exponents) than the computer can store, the result is rounded or chopped before storing.

→ d. Manual error

These errors are generated due to man made mistakes such as inaccurate data entry, inaccurate choice of numerical method, incorrect logic in program etc.

These errors can be perfectly eliminated.

e.g. Typographical error :

Instead of entering $x = 2.457892$

one may enter 24.57892

Syllabus Topic : Truncation Error and Taylor Series

1.2.2 Truncation Errors and Taylor Series

Review Question

Q. Explain truncation error and Taylor Series

- Truncation errors are those that result from using an approximation in place of an exact mathematic procedure.
- These are the errors which are resulted due to representation of infinite quantity by finite value.

For e.g : consider $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^{n-1}}{(n-1)!}$

But for calculating the value of e^x we consider only first four terms of the series. This means, we truncate the series after first four terms. Thus when we neglect higher values we land into truncation error

$$\text{Therefore, } e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$$

$$\text{Thus, Truncation error} = e^x - \left[1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \right]$$

- Truncation errors can be minimized by increasing the number of arithmetic operations for example, in numerical integration the truncation error can be reduced by increasing the number of step size. As the number of strips increases the error reduces.

1.2.3 Algorithmic Errors

- These errors are induced due to error in algorithm. For example, if we want to calculate

$$c = \frac{n!}{(n-r)! r!}$$

and if we define $n_1 = n!$

$$n_2 = (n-r)!$$

$$n_3 = r!$$

$$\text{then } c = \frac{n_1}{(n_2 \times n_3)}$$

- If we try to run the above formula with the help of a computer program, we won't be able to calculate $12!$ if $n > 12$. This is because the algorithm is wrong.

Syllabus Topic : True error

1.2.4 Absolute Errors

Review Question

Q. Explain absolute errors.

We know that error can be calculated as,

$$\text{Error} = \text{True or actual value} - \text{Approximate value}$$

The above error is also called **true error**. Always the magnitude of the error is considered. Then it is called absolute error i.e.

$$\text{Absolute error} = |\text{True or actual value} - \text{Approximate value}|$$

The absolute error depends upon magnitudes of actual and approximate values. Hence only absolute error does not provide complete information.

1.2.5 Relative Errors

Review Question

Q. Explain relative error.

$$\text{Relative error} = \frac{|\text{True (actual) value} - \text{Approximate value}|}{|\text{True value}|}$$

$$= \frac{|\text{Absolute error}|}{|\text{True value}|}$$

The relative error is also expressed in percentage i.e.

$$\text{Percentage relative error } (\epsilon_r) = \frac{|\text{Absolute error}|}{|\text{True value}|} \times 100$$

1.2.6 Inherent Error

Review Question

Q. Explain inherent error.

It is an error which is present in the statement of the problem before its solution. Thus inherent error is out of control. Inherent error can only be reduced by defining the problem correctly.

e.g. for simple pendulum equation can be written as,

$$I \frac{d^2\phi}{dt^2} + \mu \frac{d\phi}{dt} + g \sin \phi = 0$$

where it was assumed (air friction) is proportional to velocity. Without these assumption, the equation can't be derived.

1. Absolute error in summation and subtraction

Let us define the absolute error as ϵ_a . If two numbers are added or subtracted, then the magnitude of absolute error in the result is equal to sum of magnitudes of individual absolute errors.

2. Absolute error in Product

Let there be two numbers with actual values 'a' and 'b'. Let the absolute error in 'a' is ϵ_1 and that in 'b' is ϵ_2 .

$$\therefore \text{Approximate value of 'a'} = a + \epsilon_1$$

$$\text{and Approximate value of 'b'} = b + \epsilon_2$$



$$\left(\begin{array}{c} \text{Error in the} \\ \text{product} \end{array} \right) = \left[\left(\begin{array}{c} \text{Approximate} \\ \text{value of 'a'} \end{array} \right) \times \left(\begin{array}{c} \text{Approximate} \\ \text{value of 'b'} \end{array} \right) \right] - \left[\left(\begin{array}{c} \text{Actual value of 'a'} \end{array} \right) \times \left(\begin{array}{c} \text{Actual} \\ \text{value of 'b'} \end{array} \right) \right]$$

$$\begin{aligned} \epsilon &= [(a + \epsilon_1)(b + \epsilon_2)] - (ab) \\ &= ab + a\epsilon_2 + b\epsilon_1 + \epsilon_1\epsilon_2 - ab \\ &= a\epsilon_2 + b\epsilon_1 + \epsilon_1\epsilon_2 \end{aligned}$$

Since $\epsilon_1 \epsilon_2$ is very small product, it can be neglected.
Hence, absolute error in the product is given as :

$$\epsilon = a\epsilon_2 + b\epsilon_1$$

3. Absolute error in the division

Here we take the division of the two numbers, i.e.

$$\therefore \left(\begin{array}{c} \text{Error in the} \\ \text{division} \end{array} \right) = \left[\frac{\text{Approximate value of 'a'}}{\text{Approximate value of 'b'}} \right] - \left[\frac{\text{Actual value of 'a'}}{\text{Actual value of 'b'}} \right]$$

$$\text{i.e. } \epsilon_a = \frac{a + \epsilon_1}{b + \epsilon_2} - \frac{a}{b} = \frac{b\epsilon_1 - a\epsilon_2}{b \left(1 + \frac{\epsilon_2}{b} \right)}$$

Since $\frac{\epsilon_2}{b}$ is very small number compared to 1, it can be neglected. Hence,

$$\epsilon_a = \frac{b\epsilon_1 - a\epsilon_2}{b}$$

Example 1.2.1

Round the following numbers to two decimal places :

- (i) 58.22421 (ii) 5.475
(iii) 5.485 (iv) 92.3742

Solution :

- (i) 58.22421 $\Rightarrow$ 58.22
 ↑
 less than 5 (half unit)
(ii) 5.475 $\Rightarrow$ 5.47
 ↑
 > 5, and '7' is odd
(iii) 5.485 $\Rightarrow$ 5.48
 ↑
 > 5, but '8' is even
(iv) 92.3742 $\Rightarrow$ 92.37
 ↑
 > 5

Example 1.2.2

Explain the term significant figure. Round the following numbers to two decimal places.

- (i) 24.5431 (ii) 7.4679 (iii) 102.654

What would be the effect of truncating the above numbers ?

Solution :

The term significant figures are discussed earlier in the section 1.1.

- (i) Rounding off : 24.5431 $\approx$ 24.54
 Truncation : 24.5431 $\approx$ 24.54

- (ii) Rounding off : 7.4679 $\approx$ 7.47
 Truncation : 7.4679 $\approx$ 7.46
(iii) Rounding off : 102.654 $\approx$ 102.65
 Truncation : 102.654 $\approx$ 102.65

Example 1.2.3

(a) Explain the term significant figure. Round the following numbers to two decimal places.

- (i) 48.2141 (ii) 2.375 (iii) 81.255.

(b) Round off the following numbers to four significant figures,

- (i) 38.46235 (ii) 0.70029 (iii) 0.0022218

Solution :

(a) The term significance figures are discussed earlier in the section 1.1.

- (i) 48.2141 $\approx$ 48.21 (ii) 2.375 $\approx$ 2.38

- (iii) 81.255 $\approx$ 81.26

(b) (i) 38.46235 $\approx$ 38.4624 (ii) 0.70029 $\approx$ 0.7003

- (iii) 0.0022218 $\approx$ 0.0022

Example 1.2.4

If $x = 3.4535$ is correct to 4 significant figures, find the relative error.

Solution :

$x = 3.4535$ is correct to 4 digits, hence

$$3.4535 - 0.0005 \leq x \leq 3.4535 + 0.0005$$

$$\text{i.e. } 3.4530 \leq x \leq 3.4540$$

$$\therefore \text{Relative error} = \frac{3.454 - 3.453}{3.454} \times 100$$

$$= 0.028\% \quad \dots \text{Ans.}$$

Example 1.2.5

If $x = 3.4327$ find the values of absolute and relative errors if

- (i) x is truncated to three decimal places.
(ii) x is rounded off to three decimal places

Solution : The true value of $x = 3.4327$

(i) If x is truncated we get,

$$x_a = 3.4327$$

$$\therefore \text{Error} = \epsilon_x = 3.4327 - 3.432 = 0.0007$$

$$\begin{aligned} \text{Now relative error } \epsilon_r &= \frac{\epsilon_x}{x} = \frac{0.0007}{3.4327} \times 100\% \\ &= 0.0002039 \times 100\% \\ &= 0.020\% \end{aligned}$$

(ii) If x is rounded we get,

$$x_a = 3.433$$

$$\therefore \text{error, } \epsilon_x = x - x_a = -0.0003$$

$$\therefore \text{Relative error, } \epsilon_r = \frac{\epsilon_x}{x} = \frac{-0.0003}{3.4327} \times 100\%$$

$$\therefore \epsilon_r = -8.7394 \times 10^{-3} \% \quad \dots \text{Ans.}$$

Example 1.2.6

Current flows through a 10 ohm resistance that is accurate within 10%. The current is measured as 2.0 A within ± 0.1 Amps. What are absolute and relative errors in the compound voltage ? Neglect round off errors.

Solution :

$R = 10 \Omega$ within accuracy of 10%

$I = 2 \text{ A}$ within $\pm 0.1 \text{ Amps}$.

Case 1 : If R is measured 10% more then,

$$R = 10 + \frac{10}{10} = 11$$

and I is measured with 0.1 A then,

$$I = 2.1$$

$$\therefore V = IR = 11 \times 2.1 = 23.1$$

$$\text{But } V_{\text{actual}} = 10 \times 2 = 20$$

$$\therefore \text{Error} = 23.1 - 20 = 3.1 \text{ V}$$

$$\therefore \text{Relative error} = 0.155$$

Case 2 : If R is measured 10% less then,

$$R = 10 - \frac{10}{10} = 9$$

and I is measured with 0.1 A then,

$$\therefore V = 9 \times 2.1 = 18.9$$

$$\text{But } V_{\text{actual}} = 20$$

$$\therefore \text{error} = 18.9 - 20 = -1.1$$

$$\therefore \text{Relative error} = 0.055$$

Case 3 : If R is measured 10% more then,

$$R = 10 + \frac{10}{10} = 11$$

If I is measured with -0.1 A then,

$$I = 1.9$$

$$\therefore V = IR = 11 \times 1.9 = 20.9$$

$$\text{But } V_{\text{actual}} = 20$$

$$\therefore \text{error} = 20.9 - 20 = 0.9$$

$$\therefore \text{Relative error} = 0.045$$

Case 4 : If R is measured 10% less then,

$$R = 10 - \frac{10}{10} = 9$$

If I is measured with -0.1 A then,

$$I = 1.9$$

$$\therefore V = 9 \times 1.9 = 17.1$$

$$\text{But } V_{\text{actual}} = 20$$

$$\therefore \text{error} = 17.1 - 20 = -2.9$$

$$\therefore \text{Relative error} = -0.145$$

...Ans.

Example 1.2.7

The volume of a cone having base circle radius r and height h is calculated as $V = \frac{\pi}{3} r^2 h$. If $r = 3.5 \pm 0.01$ and $h = 12.5 \pm 0.01$, then compute the absolute accuracy.

Solution :

Given : $r = 3.5$, $\Delta r = 0.01$, $h = 12.5$, $\Delta h = 0.01$

Now for volume

$$\Delta v = \frac{\partial v}{\partial r} \Delta r + \frac{\partial v}{\partial h} \Delta h \quad \dots(1)$$

$$\text{As } V = \frac{\pi}{3} r^2 h$$

$$\frac{\partial v}{\partial r} = \frac{2\pi rh}{3}$$

$$\frac{\partial v}{\partial h} = \frac{\pi}{3} r^2$$

Substituting in Equation (1), we get,

$$\Delta v = \frac{2\pi rh}{3} \Delta r + \frac{\pi}{3} r^2 \Delta h$$

$$\text{As } r = 3.5 \quad \text{and } h = 12.5$$

$$\Delta r = 0.01; \quad \Delta h = 0.01$$

$$\Delta v = \frac{2 \times 3.5 \times 12.5 \pi}{3} \times 0.01 + \frac{\pi}{3} (3.5)^2 \times 0.01$$

$$\Delta v = 1.044579$$

Example 1.2.8

Suppose that you have the task of measuring the lengths of bridge and rivet and come up with 9999 and 9 cm respectively. If the true values are 10,000 and 10 cm respectively, then compute,

- Error,
- Percentage relative error in each case.

Solution :

True value of bridge = 10000 cm Errored value = 9999 cm

True value of rivet = 10 cm Errored value = 9 cm

(i) To find error

Error in bridge will be = True value - Errored value = 10000 - 9999 = 1 cm

Similarly error in the rivet will be = True value - Errored value = 10 - 9 = 1 cm

(ii) To find percentage relative error

1. Percentage error for bridge =

$$\frac{|\text{True value} - \text{errored value}|}{|\text{True value}|} \times 100 = \frac{1}{10,000} \times 100 = 0.01 \%$$

2. Percentage error for rivet =

$$\frac{|\text{True value} - \text{errored value}|}{|\text{True value}|} \times 100 = \frac{1}{10} \times 100 = 10 \%$$

Note : Even though the absolute error in both the measurements is same, percentage error for rivet is much larger than the percentage error for bridge.

Example 1.2.9

Find the error E_x and the relative error R_x and also determine the number of significant digits in the approximation for the following :

- Let $x = 2.71828182$ and $\bar{x} = 2.7182$

(ii) Let $y = 98.350$ and $\bar{y} = 98.000$

(iii) Let $z = 0.000068$ and $\bar{z} = 0.00006$

Solution :

(i) $x = 2.71828182$

$\bar{x} = 2.7182$

$E_x = 2.71828182 - 2.7182 = 0.00008182$

$\therefore R_x = \frac{0.00008182}{2.71828182} \times 100\% = 0.003\%$

(ii) $y = 98.350$

$\bar{y} = 98.000$

$E_y = 98.350 - 98.000 = 0.350$

$\therefore R_y = \frac{0.350}{98.350} \times 100\% = 0.3558\%$

(iii) $z = 0.000068$

$\bar{z} = 0.00006$

$E_z = 0.000068 - 0.00006 = 0.000008$

$\therefore R_z = \frac{0.000008}{0.000068} \times 100\% = 11.76\%$

Errors when induced in further calculations, induce further errors. This is called propagation of errors.

Example 1.2.10

Round off the number 6.65250 to four significant figures and compute absolute, relative and percentage error.

Solution :

The true value = 6.65250 (x)

Round off value = 6.652 ($\bar{x}$)

(i) Absolute error $\epsilon_x = x - \bar{x} = 0.0005$...Ans.

(ii) Relative error $= \epsilon_r = \frac{\epsilon_x}{x} = \frac{0.0005}{6.65250}$

$= 75.159 \times 10^{-6}$...Ans.

(iii) % Relative error $= \epsilon_r \times 100$

$= 75.159 \times 10^{-6} \times 100$

$= 0.0075159$...Ans.

Example 1.2.11

Volume of a certain solid V is calculated using formula, $V = \frac{6.4xy^4}{z^2}$ where x , y and z denote three dimensions. If maximum possible error in measurement of x , y and z is limited to plus minus 0.001. Estimate the maximum probable error in calculation of volume if the nominal dimension x , y and z are equal to unity.

Solution : We can write error in volume as,

$$\Delta V = \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y + \frac{\partial f}{\partial z} \Delta z$$

$$\therefore \Delta V = \frac{6.4y^4}{z^2} \epsilon_x + \frac{25.6xy^3}{z^2} \epsilon_y - \frac{12.8xy^4}{z^3} \epsilon_z$$

As $x = y = z = 1$

$\therefore \Delta V = 64 \times (\pm 0.001) + 25.6 (\pm 0.001) - 12.8 (\pm 0.001)$

To get maximum possible error, we take,

$\epsilon_x = \epsilon_y = 0.001$ and $\epsilon_z = -0.001$

$\therefore \Delta V = 64(0.001) + 25.6(0.001) - 12.8(-0.001)$

$\therefore \Delta V = 0.0448$

Thus maximum error induced is

$\Delta V = 0.0448$

...Ans.

Example 1.2.12

To find the approximate maximum error in 5.43×27.2 .

Solution : The magnitude of the maximum possible error in 5.43 is 0.005 and in 27.2 is 0.05.

Hence the approximate maximum error in 5.43×27.2 is

$0.005 \times 27.2 + 0.05 \times 5.43 = 0.4$

Now, the actual product is 147.7 and it can be expected to lie between 147.3 and 148.1.

To obtain a formula giving the error in quotient.

$$\begin{aligned} \text{Error} &= \frac{\epsilon_1}{\epsilon_2} - \frac{X_1}{X_2} = \frac{X_1 + \epsilon_1}{X_2 + \epsilon_2} - \frac{X_1}{X_2} \\ &= \frac{\epsilon_1 X_2 - \epsilon_2 X_1}{X_2(X_2 + \epsilon_2)} = \frac{\epsilon_1 X_2 - \epsilon_2 X_1}{X_2^2 \left[1 + \frac{\epsilon_2}{X_2} \right]} \end{aligned}$$

If ϵ_2 is small in comparison with X_2 , then $\frac{\epsilon_2}{X_2}$ is small

compared to 1, then the error is approximate.

$$\frac{\epsilon_1 X_2 - \epsilon_2 X_1}{X_2^2} = \left[\frac{\epsilon_1}{X_1} - \frac{\epsilon_2}{X_2} \right] \frac{X_1}{X_2}$$

Example 1.2.13

The quotient $\frac{25.4}{12.37}$ gives the result 2.053. To what extent can this be relied upon?

Solution : The magnitude of the maximum possible error in 25.4 is 0.05 and in 12.37 is 0.005.

Hence an estimate of the maximum possible error in the quotient is

$$\left[\frac{0.05}{25.4} - \frac{0.005}{12.37} \right] \frac{25.4}{12.37} = 0.005$$

Hence the true quotient can be expected to lie between 2.048 and 2.058.

Note : It may be assumed that $\frac{x_1}{x_2}$ and $x_1 x_2$ are not much different from $\frac{X_1}{X_2}$ and $X_1 X_2$.

Example 1.2.14

Find the value of $\sin x$, using series expansion

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

for $x = 0.2$, with absolute error smaller than 0.005.

Solution :

Step 1 : Recall the expansion of $\sin x$;

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{(-1)^n x^{2n-1}}{(2n-1)!}$$



the n^{th} term = $\frac{(-1)^n x^{2n-1}}{(2n-1)!}$

If we use n terms of the series to find $\sin x$, then the magnitude of the truncated error is

$$\left| \frac{(-1)^{n+1} x^{2n+1}}{(2n+1)!} \right| = \frac{x^{2n+1}}{(2n+1)!}$$

In order that error be less than 0.005, we write,

$$\frac{x^{2n+1}}{(2n+1)!} < 0.005$$

$$\therefore (2n+1) \log x - \log (2n+1)! < \log 0.005$$

$$(2n+1) \log x - \log (2n+1)! < -2.30$$

$$\text{i.e. } \log (2n+1)! - (2n+1) \log x < 2.30$$

Step 2 : Now for $n=1$; $x=0.2$;

$$\log (2n+1)! - (2n+1) \log x = \log 6 - 3 \log 0.2 = 2.8750$$

and $2.8750 < 2.30$

Hence for absolute error smaller than 0.005; we take $n=1$;

$$\text{i.e. } \sin x = x$$

Hence, **at $x = 0.2$, $\sin x = 2.875$Ans.**

Example 1.2.15

Find the errors introduced when (a) 18.496, (b) 18.493, (c) 17.208 are rounded by one digit. Then find (d) 18.496 + 18.493 (e) 18.496 + 17.208, (f) 18.496 - 18.493 (g) 18.496 - 17.208 and the errors introduced when (a), (b), (c) are first rounded by one digit and then the sums and differences corresponding to (d), (e), (f) and (g) are formed.

Solution :

- (a) 18.496 becomes 18.50 introducing error - 0.004
- (b) 18.493 becomes 18.49 introducing error - 0.003
- (c) 17.208 becomes 17.21 introducing error - 0.002.
- (d) 36.989.
- (e) 35.704
- (f) 0.003
- (g) 1.288

$$18.50 + 18.49 = 36.99 \text{ introducing error } - 0.001$$

$$18.50 + 17.21 = 35.71 \text{ introducing error } - 0.006$$

$$18.50 - 18.49 = 0.01 \text{ introducing error } - 0.007$$

$$18.50 - 17.21 = 1.29 \text{ introducing error } - 0.002.$$

Example 1.2.16

Evaluate the following expressions. Then, assuming maximum round-off error in each figure, estimate the limits between which your calculated value probably lies. Finally, quote each result to as many significant figures as are reliable.

(a) $\frac{2.362 \times 1.76}{1.46 \times 0.785}$ (b) $7.62 + \frac{54}{15.3}$ (c) $\frac{1.32 - 0.463}{16.1 \times 2.17}$

In (b) assume that there is no error in the number 54.

Solution :

(a) Numerator = 4.157

$$\text{Estimate of error in numerator} = (0.0005 \times 1.76) + (0.005 \times 2.362)$$

$$= 0.0009 + 0.012 = 0.0129$$

$$\text{Denominator} = 1.146$$

$$\text{Estimate of error in denominator} = (0.005 \times 0.785) + (0.0005 \times 1.46)$$

$$= 0.004 + 0.0007 = 0.0047$$

$$\text{Estimate of final error} = \left(\frac{0.0129}{4.157} + \frac{0.0047}{1.146} \right) \frac{4.157}{1.146}$$

$$= (0.003 + 0.004) 3.6 = 0.025$$

$$= \text{Limits : } 3.602 \text{ to } 3.652.$$

Actual answer is 3.627.

(b) Estimate of error in quotient = $\left(\frac{0.05}{15.3} \right) \frac{54}{15.3} = 0.012$

$$\text{Estimate of final error} = 0.005 + 0.012 = 0.02$$

$$\text{Limits : } 11.13 \text{ to } 11.17$$

Actual answer is 11.15.

(c) Estimate of error in difference = $0.005 + 0.0005 = 0.0055$

$$\text{estimate of error in product} = 0.05 \times 2.17 + 0.005 \times 16.1 = 0.2$$

$$\text{Numerator} = 0.857$$

$$\text{Denominator} = 34.94$$

$$\text{Estimate of final error} = \left(\frac{0.0055}{0.857} + \frac{0.20}{34.94} \right) \frac{0.857}{34.94} = 2.91 \times 10^{-4}$$

$$\text{Limits : } 0.0242 \text{ to } 0.0248.$$

$$\text{Actual answer is } 0.0245. \quad \dots A$$

Example 1.2.17

Evaluate the following expressions for the given values of first putting each in an appropriate form for the calculation you are using.

(a) $x^2 - 2 + \frac{1}{x^2}$ when $x = 1.473$

(b) $x^4 - 3.17x^3 + 5.29x - 17.3$ when $x = 5.12$.

Estimate the approximate maximum error in each result assuming all integer coefficients are exact but that decimals are subjected to the maximum round-off error.

Solution :

(a) We write $x^2 - 2 + \frac{1}{x^2} = \frac{x^4 - 2x^2 + 1}{x^2}$
 $= \frac{(x^2 - 1)^2}{x^2}$ is an appropriate form.

$$\text{At } x = 1.473; \text{ the value is } 0.631.$$

$$\text{Error in } x^2 = 2 \times 1.473 \times 0.0005 = 0.0015$$

$$\text{Error in } x^2 - 1 = 0.0015.$$

(This is the same as that in x^2 as the -1 is exact).

$$\text{Denominator} = 2.170$$

$$\therefore x^2 - 1 = 1.170$$

$$\text{Numerator} = 1.369$$

$$\therefore \text{Estimate of error in numerator} = 2 \times 1.369 \times 0.0015 = 0.0041.$$

Estimate of error in final result

$$= \left(\frac{0.0041}{1.369} + \frac{0.0015}{2.170} \right) \frac{1.369}{2.170} = 0.002$$

(b) We write the given form as $\{(x - 3.17)x^2 + 5.29\}x - 17.3$

$$\text{When } x = 5.12; \text{ the value is } 287.1.$$

Approximate maximum error in

$$x^2 = 2 \times 5.12 \times 0.005 = 0.051$$

$$x^2 = 26.21$$

$$x - 3.17 = 1.95$$

$$\therefore (x - 3.17)x^2 + 5.29 = 51.12$$

Approximate error in $(x - 3.17) = 0.005 + 0.005 = 0.01$.

Approximate error in $(x - 3.17) x^2 = 0.01 \times 26.21 + 1.95 \times 0.051 = 0.451$

Approximate error in $(x - 3.17) x^2 + 5.29 = 0.451 + 0.005 = 0.456$

Approximate error in

$$\{(x - 3.17) x^2 + 5.29\} x = 0.456 \times 5.12 + 0.005 \times 51.12 = 2.6$$

Note that the subtraction of 1.73 will not affect this error to the degree of accuracy quoted.

Example 1.2.18

Find as accurately as your tables allow,

$\log [x - \sqrt{x^2 - 1}]$ when $x = 27.3$

Solution :

$$\text{We write } \log [x - \sqrt{x^2 - 1}] = \log \frac{1}{x + \sqrt{x^2 - 1}}$$

$$\text{when } x = 27.3, \quad \text{Expn} = \log \frac{1}{54.59} = -\log 54.59 = -3.998$$

$$= -4.00 \quad \dots \text{Ans.}$$

Example 1.2.19

Find the smaller root of the equation $x^2 - 400x + 1 = 0$ using four digit arithmetic.

Solution : The result for the smaller root is given by,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \dots (i)$$

$$\text{Here } a = 1 = 0.1000 \times 10^1$$

$$b = 400 = 0.4000 \times 10^3$$

$$c = 1 = 0.1000 \times 10^1$$

$$\therefore b^2 - 4ac = (0.1600 \times 10^6) - (0.4000 \times 10^1)$$

$$= 0.1600 \times 10^6 \text{ (to four digit accuracy)}$$

$$\therefore \sqrt{b^2 - 4ac} = 0.4000 \times 10^3$$

Substituting in the Equation (i), we get $x = 0.0000$

But, if we rewrite the above formula in the form

$$x = \frac{[b - \sqrt{b^2 - 4ac}][b + \sqrt{b^2 - 4ac}]}{2a[b + \sqrt{b^2 - 4ac}]} = \frac{2c}{b + \sqrt{b^2 - 4ac}}$$

Then we get,

$$x = \frac{0.2000 \times 10^1}{0.4000 \times 10^3 + 0.4000 \times 10^3} = \frac{0.2000 \times 10^1}{0.8000 \times 10^3}$$

$$= 0.0025 \text{ which is the exact root.} \quad \dots \text{Ans.}$$

Example 1.2.20

Compute the middle value of the numbers $a = 4.568$ and $b = 6.762$ using the four digit arithmetic.

Solution : If we take the middle value as the mean of the numbers, we write

$$c = \frac{a+b}{2} = \frac{(0.4568 \times 10^1) + (0.6762 \times 10^1)}{2} = 0.5660 \times 10^1$$

But, if we use the formula,

$$c = a \frac{b-a}{2}$$

$$\text{We get, } c = 0.4568 \times 10^1 + \frac{0.6762 \times 10^1 - 0.4568 \times 10^1}{2}$$

$$= 0.5665 \times 10^1 \text{ which is the accurate result.} \quad \dots \text{Ans.}$$

Example 1.2.21

In the evaluation of the expression

$$x^4 - 3.18 x^3 + 5.36 x - 1.187 \text{ at } x = 5.29.$$

Estimate the maximum error assuming that all decimals are subjected to a maximum rounding off error.

Solution :

$$f = x^4 - 3.18 x^3 + 5.36 x - 1.187$$

We have to calculate the maximum error at $x = 5.29$

$$x^4 = (5.29)^4 = 783.10985$$

$$3.18 \times x^3 = 3.18 \times (5.29)^3 = 470.75413$$

$$5.36 \times x = 5.36 \times 5.29 = 28.3544$$

$$\therefore f_1 = 783.10985 - 470.75413 + 28.3544 - 1.187$$

$$\therefore f_1 = 339.52312$$

Now if the values are rounded off to 3 decimal places then,

$$x^4 = (5.29)^4 = 783.110$$

$$3.18 x^3 = 3.18 \times (5.29)^3 = 470.754$$

$$5.36 \times x = 5.36 \times 5.29 = 28.354$$

$$\therefore f_2 = 783.110 - 470.754 + 28.354 - 1.187$$

$$\therefore f_2 = 339.523$$

$$\therefore \text{Error} = f_1 - f_2 = 0.00012 \quad \dots \text{Ans.}$$

Example 1.2.22

Obtain the value of expression $(1+x)^2$ by two methods using four digit floating point arithmetic for $x = 0.5129$. Calculate the relative errors in two methods.

Solution : The expression $(1+x)^2$ can be expanded as :

$$(i) (1+x)^2 = (1+x) \cdot (1+x)$$

$$\text{and } (ii) (1+x)^2 = 1 + 2x + x^2$$

1. To calculate exact value of $(1+x)^2$

$$\text{Now, } (1+x)^2 = (1+0.5129)^2 = 2.2888664 = 2.2888664 \times 10^1$$

2. To calculate the relative error in $(1+x)^2 = (1+x) \cdot (1+x)$

$$\text{We write, } (1+x)^2 = (1+0.5129)(1+0.5129)$$

$$= (1.5129)(1.5129)$$

The number 1.5129 can be represented by four digit floating point arithmetic.

It is 0.1512,

$$\therefore (1+x)^2 = (0.1512)(0.1512)$$

$$= 0.0228614 \text{ (floating point multiplication)}$$

$$= 0.0228 \text{ (Four digits mantissa)}$$

$$= 0.2280 \text{ (Normalised representation)}$$

$$= 0.2280 \times 10^1$$

$$\text{Hence, relative error} = \frac{2.2888664 \times 10^1 - 0.2280 \times 10^1}{0.22888664 \times 10^1}$$

$$= 3.8735339 \times 10^{-3}$$

3. Now, we calculate the relative error in $(1+x)^2 = 1 + 2x + x^2$

$$\text{Putting } x = 0.5129$$

$$\text{We have, } (1+x)^2 = 1 + (2 \times 0.5129) + (0.5129 \times 0.5129)$$

We convert the numbers in above equation to four-digit floating point arithmetic i.e.,

$$1 \rightarrow 0.1000$$

$$2 \rightarrow 0.2000$$

$$0.5129 \rightarrow 0.5129$$

Hence, the expression can be written as,

$$(1+x)^2 \rightarrow (0.1000) + (0.2000 \times (0.5129)) + (0.5129 \times 0.5129) \\ \rightarrow (0.1000) + (0.1025) + (0.2630)$$

We equalize exponents to perform the addition,

$$= (0.1000) + (0.1025) + (0.2630) = 0.2288 \\ = 0.2288 \times 10^1$$

Now relative error will be,

$$= \frac{0.22888664 \times 10^1 - 0.2288 \times 10^1}{0.22888664 \times 10^1} = 3.783533 \times 10^{-4}$$

We observe that the relative error in second method is less than the first method.

Example 1.2.23

Calculate error in the volume V of a tank is given by, $V = \pi/4 d^2 l$ at $d = 1.0$ m and $l = 2.0$ m. If error in measurement of diameter d is ± 0.01 m and l is 0.01 m.

Solution : We can write error in volume as,

$$\Delta V = \frac{\partial V}{\partial d} \cdot \Delta d + \frac{\partial V}{\partial l} \cdot \Delta l \quad \dots(1)$$

$$\text{As } V = \pi/4 d^2 l$$

$$\therefore \frac{\partial V}{\partial d} = \frac{2\pi d l}{4} = \frac{\pi d l}{2} \quad \text{and} \quad \frac{\partial V}{\partial l} = \frac{\pi}{4} \cdot d^2$$

Substituting in Equation (1) we get,

$$\Delta V = \frac{\pi d l}{2} \times \Delta d + \frac{\pi}{4} \cdot d^2 \times \Delta l$$

$$\text{As } d = 1 \text{ m and } l = 2 \text{ m and } \Delta d = 0.01 \text{ and } \Delta l = 0.01$$

$$\text{We get, } \Delta V = \frac{\pi \times 1 \times 2}{2} \times 0.01 + \frac{\pi}{4} \times 1^2 \times 0.01 \\ = 0.03926 \text{ m} \\ \Delta V = 0.03926 \text{ m} \quad \dots \text{Ans.}$$

Example 1.2.24

If $u = 3v^7 - 6v$, find the percentage of error in 'u' and $v = 1$, if error in 'v' is 0.05.

$$\text{Solution : } u = 3v^7 - 6v \quad \therefore \Delta u = \frac{\partial u}{\partial v} \times \Delta v$$

$$\text{As } \frac{\partial u}{\partial v} = 21v^6 - 6 \quad \therefore \Delta u = (21v^6 - 6) \Delta v$$

$$\text{and } \Delta v = 0.05, v = 1 \quad \dots(\text{Given})$$

$$\therefore \Delta u = (21 \times (1)^6 - 6) \times 0.05 \Delta u = 0.75$$

$$\text{and } u = 3 \times (1)^7 - 6(1) = -3$$

$$\% \text{ error} = \left| \frac{\text{True value} - \text{errored value}}{\text{True value}} \right| \times 100 \\ = \left| \frac{-3 - (-2.25)}{-3} \right| \times 100 = \left| \frac{0.75}{-3} \right| \times 100 = 25 \%$$

Example 1.2.25

Volume of cylinder is calculated after measuring its diameter (2.5 ± 0.02) m and its height as (4.8 ± 0.05) m respectively. Estimate the absolute error in calculation of volume.

Solution :

$$\text{Given data : } V = \frac{\pi}{4} d^2 \cdot l, d_1 = 2.5 \text{ m}, l_1 = 4.8 \text{ m}, \Delta d = 0.02 \text{ m},$$

$$\Delta l = 0.05 \text{ m}$$

Find : Absolute error in volume (ΔV)

Step 1 : Calculate the partial derivative of V w.r.t. diameter and length.

$$V = \frac{\pi}{4} d^2 \cdot l$$

Partially differentiating w.r.t. diameter

$$\frac{\partial V}{\partial d} = \frac{\pi}{4} 2d \cdot \frac{\partial d}{\partial d} \cdot l \quad ; \quad \frac{\partial V}{\partial d} = \frac{\pi d l}{2}$$

Partially differentiating w.r.t. length

$$\frac{\partial V}{\partial l} = \frac{\pi}{4} d^2 \frac{\partial l}{\partial l} \quad ; \quad \frac{\partial V}{\partial l} = \frac{\pi d^2}{4}$$

Step 2 : Calculate the absolute error in volume

$$\Delta V = \left| \frac{\partial V}{\partial d} \cdot \Delta d \right| + \left| \frac{\partial V}{\partial l} \cdot \Delta l \right|$$

$$\Delta V = \left| \frac{\pi d l}{2} \cdot \Delta d \right| + \left| \frac{\pi d^2}{4} \cdot \Delta l \right|$$

$$= \left| \frac{\pi \times 2.5 \times 4.8}{2} \times 0.02 \right| + \left| \frac{\pi \times 2.5^2}{4} \times 0.05 \right|$$

$$\therefore \Delta V = 0.3769 + 0.2454 \quad \therefore \Delta V = 0.6223 \text{ m}^3 \quad \dots \text{Ans.}$$

Syllabus Topic : Precision and Accuracy

1.3 Precision and Accuracy

Review Question

Q. Explain the accuracy and precision.

- **Accuracy :** It refers to how closely a computed or measured value agrees with the true value.
- **Precision :** It refers to how closely an individual computed or measured values agree with each other.
- The numerical solutions should be sufficiently accurate to meet the requirements of particular engineering problem. They should also be precise enough for adequate engineering design.

Syllabus Topic : Machine epsilon

1.4 Machine Epsilon

Machine epsilon, ϵ_{mach} is defined as the smallest number such that $1 + \epsilon_{\text{mach}} > 1$. It is difference between 1 and the next nearest number representable as a machine number,

For example : The number is 1.0625

$$\text{Then } \epsilon_{\text{mach}} = 1.0625 - 1 = 0.0625$$

Syllabus Topic : Data Uncertainties

1.5 Data Uncertainties

Uncertainty is an inherent part of analysis. Errors enters into an analysis due to uncertainty in measurement, experiments, simulation (i.e. physical data) on which model is based. For instant, supposed we conduct an experiment using bomb calorimeter to find heating value of one gram of sample. The experiments are able to find the heating value of a one gram sample with uncertainty less than two percent. However, there is a significant amount of variation within a sample of lignite and even more variation between different samples of lignite. Thus, the uncertainty arises

not from the measurement technique but rather from the variation within the physical variable itself.

Uncertainty of data traditionally focuses on the accuracy of a particular measurement. However, it can also be used to investigate the entire range of potential/physical values which may exist for a parameter.

Syllabus Topic : Error Propagation

1.6 Error Propagation

Review Questions

- Q. Explain error propagation.
Q. Write a short note on propagation of errors.

Error propagates during calculations when the next approximation is based on the previous approximation. Let us consider how error propagates when there is addition, subtraction, multiplication, division and exponentiation.

Case 1 : Addition

$$\text{Let } z = x + y$$

$$\text{Let error in } x \quad \epsilon_x = x - x_a$$

$$\text{Let error in } y \quad \epsilon_y = y - y_a$$

$$\text{Let error in } z \quad \epsilon_z = z - z_a$$

where the suffix 'a' represents approximate value

$$\therefore z - z_a = (x + y) - (x_a + y_a) = x - x_a + y - y_a$$

$$\therefore z - z_a = \epsilon_x + \epsilon_y$$

Thus if $\epsilon_x, \epsilon_y > 0$ then error will increase whereas if $\epsilon_x, \epsilon_y < 0$ error will decrease.

Case 2 : Subtraction

$$\text{Let } z = x - y$$

$$\text{Let error in } z \quad \epsilon_z = z - z_a$$

where the suffix 'a' represents approximate value

$$\therefore z - z_a = (x - y) - (x_a - y_a) = (x - x_a) + (y - y_a)$$

$$= \epsilon_x - \epsilon_y$$

Thus if $\epsilon_x, \epsilon_y > 0$ then error will reduce whereas if $\epsilon_x, \epsilon_y < 0$ error will increase.

Case 3 : Multiplication

$$\text{Let } z = x \times y$$

$$\epsilon_z = z - z_a; \epsilon_x = x - x_a; \epsilon_y = y - y_a$$

$$\text{Now } z - z_a = x \cdot y - x_a \cdot y_a$$

$$\therefore z - z_a = xy(1 - (x - x_a)(y - y_a))$$

$$= xy - xy - x\epsilon_y + y\epsilon_x - \epsilon_x\epsilon_y$$

$$= z \left(1 - 1 + \frac{\epsilon_y}{y} + \frac{\epsilon_x}{x} + \frac{\epsilon_x\epsilon_y}{xy} \right)$$

Now neglecting $\frac{\epsilon_x\epsilon_y}{xy}$ which induces further more error we get,

$$z - z_a = \frac{\epsilon_x}{x} + \frac{\epsilon_y}{y}$$

Thus if $\epsilon_x, \epsilon_y > 0$ then error will increase whereas if $\epsilon_x, \epsilon_y < 0$ error will reduce.

Case 4 : Division

$$\text{Let } z = \frac{x}{y}; \epsilon_z = z - z_a = \frac{x}{y} - \frac{x_a}{y_a}$$

$$\begin{aligned} \therefore \epsilon_z &= z - z_a = \frac{x}{y} - \frac{x(1 - \epsilon_x/x)}{y(1 - \epsilon_y/y)} \\ &= z \left[1 - \left(1 - \frac{\epsilon_x}{x} \right) \left(1 - \frac{\epsilon_y}{y} \right)^{-1} \right] \end{aligned}$$

is an infinite series, Now $\left(1 - \frac{\epsilon_y}{y} \right)^{-1}$

$$\left(1 - \frac{\epsilon_y}{y} \right)^{-1} = 1 + \frac{\epsilon_y}{y} + \frac{\epsilon_y^2}{y^2} + \dots$$

Neglecting higher terms thus inducing further error we get,

$$\epsilon_z = z - z_a = z \left[1 - \left(1 - \frac{\epsilon_x}{x} \right) \left(1 + \frac{\epsilon_y}{y} \right) \right]$$

$$\therefore \frac{\epsilon_z}{z} = 1 - \left[1 + \frac{\epsilon_y}{y} - \frac{\epsilon_x}{x} - \frac{\epsilon_x\epsilon_y}{xy} \right] = \frac{\epsilon_x}{x} - \frac{\epsilon_y}{y}$$

Thus if $\epsilon_x, \epsilon_y > 0$ error will reduce whereas if $\epsilon_x, \epsilon_y < 0$ error will increase.

Case 5 : Exponentiation

$$\text{Let } z = x^p \quad \therefore z_a = x_a^p$$

$$\begin{aligned} \therefore \epsilon_z &= z - z_a = x^p - (x - \epsilon_x)^p \\ &= x^p - (x^p - px^{p-1}\epsilon_x + \dots) \end{aligned}$$

Neglecting higher terms thus inducing further error we get,
 $\epsilon_z = px^{p-1} \cdot \epsilon_x$

$$\therefore \frac{\epsilon_z}{z} = \frac{px^{p-1} \cdot \epsilon_x}{x^p}$$

$$\therefore \frac{\epsilon_z}{z} = p \frac{\epsilon_x}{x}$$

Thus relative error in z is p times the relative error in x . Thus if ϵ_x reduces the error will reduce.

Syllabus Topic : Importance of Errors in Computer Programming

1.7 Importance of Errors in Computer Programming

- A program is a problem statement whose solution is to be found out. A computer program is a sequence of instructions written in some programming language to solve a particular problem by using computer.
- Programs are compiled by using compiler to check for any error in the program.
- While doing the computer program, errors are naturally encountered understanding and fixing errors is an important part of computer programming to generate accurate result.
- So programmer has to make efforts to detect and rectify any kind of errors are present in the programme but before detecting and removing errors it is much more necessary that the programmer should know about the types of errors in programming.